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SmartEdu: Optimizing Institutional Resource Allocation for Maximum Academic ROI

Project Domain

Excel / Power BI with Copilot / AI Chatbot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

SmartEdu is a data-driven Business Intelligence project developed to optimize institutional resource allocation for maximum academic Return on Investment (ROI). Educational institutions often invest heavily in resources such as teacher quality, tutoring programs, and infrastructure, yet lack a structured framework to measure the actual academic impact of these expenditures. This project addresses that gap by analyzing a dataset of 6,500+ student records sourced from Kaggle, covering 20 variables including academic performance indicators, behavioral traits, and socio-economic backgrounds. Using Microsoft Excel for data cleaning and statistical analysis, Power BI for interactive dashboard development, and an AI-powered Chatbot (Edu-Advisor) for natural language querying, the project quantifies the marginal impact of high-cost interventions on exam scores. Key objectives include calculating academic ROI, developing segmented investment strategies based on income and demographic variables, and deploying advanced visualizations. The final deliverables include a dynamic Power BI dashboard, an Investment Prioritization Matrix, and a conversational AI chatbot that enables school administrators to retrieve real-time insights without navigating complex databases. The project demonstrates a full-stack BI workflow from raw data ingestion to AI-integrated analytics, offering a replicable and scalable framework for evidence-based educational decision-making.

SmartEdu focuses on quantifying academic ROI by analyzing student performance data using Excel, Power BI, and an AI Chatbot. The dataset of 6,500+ records was cleaned using Power Query, followed by regression and correlation analysis to identify key performance drivers. An interactive Power BI dashboard was built using DAX measures for ROI tracking across school types and income brackets. An Edu-Advisor chatbot was deployed to allow school administrators to query insights via natural language, such as pass rates for low-income students with high tutoring frequency.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project.

1.1 Problem Statement

The SmartEdu project addresses the critical lack of a data-driven framework for measuring academic Return on Investment (ROI) in educational institutions. Despite significant financial investments in resources such as teacher quality, tutoring sessions, and institutional infrastructure, school administrators currently have no systematic method to evaluate the actual academic impact of these expenditures.

- Educational institutions invest heavily in resources but operate without a measurable framework to quantify the academic ROI of those investments.
- Misallocated budgets using a “one-size-fits-all” approach led to financial inefficiency, preventing institutions from identifying whether limited funds are better spent on high-cost academic interventions or low-cost environmental improvements.
- The primary stakeholders affected include school administrators, academic counsellors, and institutional budget planners who require accessible, actionable insights from student data without needing to navigate complex statistical tools or dense spreadsheets.
- Existing approaches rely on static spreadsheets and dense reports that are not interactive, predictive, or accessible to non-technical users. There is an absence of a dynamic BI system combined with a conversational AI interface that can translate raw student data into real-time, actionable insights.
- **Accessibility Gap:** Even when student data is analysed, findings are locked in static spreadsheets inaccessible to non-technical administrators. School counsellors need an intuitive interface to ask context-specific questions such as “Which students need tutoring the most this week?” without navigating complex databases.
- **Analytical Gap:** There is an immediate need to translate raw student data from 6,500+ records into an interactive visual dashboard and a conversational AI assistant, enabling institutions to strategically deploy their budgets and intervene in real-time to improve student outcomes.

1.2 Motivation

- The education sector is increasingly generating vast amounts of data, yet institutions rarely leverage it for evidence-based decision-making. This project is motivated by the goal of bridging that gap — ensuring that every rupee invested in student development is measurable, justified, and impactful.
- The SmartEdu framework can be replicated across schools, colleges, and universities to guide budget planning, identify at-risk students, and design demographically targeted intervention programs based on socio-economic and behavioural data.
- By replacing intuition-based budget allocation with a data-backed Investment Prioritization Matrix, institutions can achieve significantly higher academic yields from their existing resources and reduce financial waste on low-impact interventions.

- This project aligns with the growing industry adoption of Business Intelligence tools and conversational AI for institutional analytics. The integration of Power BI dashboards with AI Chatbots mirrors modern enterprise-grade data solutions being adopted across the education technology sector globally.

1.3 Objectives

1. To collect, understand, and model the Student Performance Factors dataset (6,500+ records, 20 variables) covering academic, behavioural, and socio-economic attributes sourced from Kaggle.
2. To clean and preprocess raw data using Excel Power Query — standardizing formats, handling missing values, and building a relational data model mapping institutional investments against student behaviours and socio-economic baselines.
3. To calculate Academic ROI by quantifying the marginal gain in exam scores from high-cost interventions (e.g., tutoring sessions, teacher quality) using Excel Analysis ToolPak for correlation and regression analysis.
4. To build an advanced Power BI dashboard with DAX measures for ROI tracking, drill-through pages for demographic analysis, and interactive visualizations enabling school boards to filter performance across school types, income brackets, and key metrics.
5. To deploy an Edu-Advisor AI Chatbot using Microsoft Copilot Studio enabling school administrators to query the dataset using natural language and receive instant data-backed answers on student risk factors and resource allocation recommendations.
6. To produce an Investment Prioritization Matrix identifying which institutional factors yield the highest academic returns, providing a replicable and scalable evidence-based framework for educational budget planning.

1.4 Scope of the Project

- This project covers data from the Student Performance Factors dataset (Kaggle), containing 6,500+ student records and 20 variables spanning academic scores, behavioral traits (hours studied, attendance, sleep), and socio-economic indicators (family income, parental education, access to resources).
- The tools used are Microsoft Excel (Power Query, Analysis ToolPak), Power BI Desktop (DAX, drill-through visualizations), Microsoft Copilot Studio / Power BI Q&A (AI Chatbot integration), and Power BI Service for publishing.
- The analysis focuses on data cleaning and relational modelling, statistical ROI calculation, interactive BI dashboard development with demographic drill-downs, and conversational AI integration for natural language querying of institutional data.
- The project does not include real-time data pipelines, predictive machine learning models, live API integrations, or mobile application development. The chatbot is configured to the provided dataset and is not a general-purpose AI assistant.
- The project is intended as a fellowship showcase and is not a production-level system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

- Levin and Belfield (2015) established a foundational framework for cost-effectiveness analysis in education, demonstrating that institutions can systematically compare the academic yield of different interventions. SmartEdu builds on this framework by operationalizing it within modern BI tools.
- Existing research on student performance prediction (e.g., Kaggle benchmarks, educational data mining papers) typically uses Python-based machine learning models. These require technical expertise unavailable to most school administrators. SmartEdu instead uses Excel and Power BI — tools already familiar to institutional staff — to achieve interpretable, actionable insights without complex ML pipelines.
- A key gap in existing literature is the absence of a combined BI dashboard and conversational AI interface tailored for educational ROI tracking. SmartEdu addresses this by integrating Power BI visualizations with a natural-language Edu-Advisor Chatbot, allowing non-technical administrators to retrieve instant, data-backed insights without any statistical knowledge.

2.2 Tools and Technologies Review

- Microsoft Excel — used in SmartEdu for initial data ingestion, Power Query-based cleaning, relational data modelling, and statistical ROI analysis via the Analysis ToolPak (correlation, regression). Excel serves as the data preparation backbone before Power BI import.
- Microsoft Power BI — the primary visualization and reporting platform for SmartEdu. Interactive dashboards with DAX measures track academic ROI, drill-through pages enable demographic-specific analysis (school type, income bracket), and slicers allow real-time filtering across 6,500+ student records. Power BI Service is used for publishing and sharing.
- Microsoft Copilot Studio — used to build the Edu-Advisor Chatbot. The bot is configured to read the SmartEdu dataset and Power BI dashboard, enabling school administrators to ask natural language questions such as “Show me the pass rate for low-income students with high tutoring” and receive instant data-backed answers.
- Power BI Q&A Feature — an embedded natural language query engine within Power BI that allows users to type questions directly into the dashboard and receive visual answers. Used alongside Copilot Studio to power the Edu-Advisor interface within the SmartEdu environment.
- GitHub — used for version control and project submission, hosting the final analytical report, cleaned dataset, and Power BI file for evaluation and reproducibility.

2.3 Gaps Identified

- Most existing educational analytics literature relies on complex Python or R-based machine learning pipelines. These approaches, while powerful, are inaccessible to school administrators who are not data scientists. SmartEdu bridges this gap by delivering equivalent insights through Excel and Power BI.
- There is limited published work on integrating conversational AI Chatbots directly with institutional education dashboards. Most Power BI implementations in education remain static reports without natural language query capabilities, leaving a significant usability gap for non-technical end users.
- The concept of Academic ROI measuring the precise score impact of specific institutional investments lacks standardized quantitative frameworks in the education sector. SmartEdu proposes a replicable ROI calculation methodology using regression analysis and DAX measures that can be adopted by institutions at scale.
- Socio-economic segmentation in educational analytics is rarely incorporated into institutional dashboards. SmartEdu's segmented investment strategy analysing how family income, parental education, and resource access interact with academic outcomes fills this gap with demographic-specific intervention recommendations.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

- Data Source / Input: Student Performance Factors dataset from Kaggle — a CSV file containing 6,500+ student records with 20 variables covering numerical (Hours Studied, Attendance, Sleep Hours, Previous Scores, Tutoring Sessions, Physical Activity, Exam Score), categorical (Gender, School Type, Family Income, Motivation Level, Teacher Quality, Parental Involvement, Peer Influence, Parental Education Level, Access to Resources), and binary (Internet Access, Extracurricular Activities, Learning Disabilities) fields.
- Data Cleaning: Raw data was imported into Excel using Power Query. Cleaning steps included removing duplicate records, handling null/missing values through mean imputation for numerical fields and mode substitution for categorical fields, standardizing text casing for categorical variables, validating score ranges (Exam Score: 0–100), and encoding binary variables consistently. A relational data model was then built mapping institutional investment variables (Teacher Quality, Tutoring Sessions) against student outcome variables (Exam Score) and socio-economic baselines (Family Income, Parental Education).
- Analysis / Processing: Statistical analysis was conducted in Excel using the Analysis ToolPak. Correlation analysis identified the strength of association between each variable and Exam Score. Regression analysis isolated the true marginal drivers of academic performance, calculating the ROI (score gain per unit of investment) for high-cost interventions such as tutoring sessions and teacher quality improvements. Segmentation analysis further grouped outcomes by income bracket and school type to enable targeted intervention recommendations.
- Visualization / Output: The cleaned and modelled data was imported into Power BI Desktop. An interactive multi-page dashboard was developed featuring DAX-powered KPI cards (average exam score, ROI by intervention type), bar and scatter charts for performance segmentation, drill-through pages for demographic breakdowns, and slicers for dynamic filtering. The Edu-Advisor Chatbot was integrated via Copilot Studio and Power BI Q&A to enable natural language querying. An Investment Prioritization Matrix was compiled as the final analytical report output.

3.2 Modules / Components Used

- Data Ingestion: Loading the raw Student Performance Factors CSV (6,500+ records, 20 variables) into Excel via Power Query, establishing data types and initial schema validation.
- Data Cleaning & Modelling: Removing duplicates, handling null values, standardizing categorical formats, validating score ranges, and constructing a relational data model in Excel that maps investment variables to academic outcomes and socio-economic baselines.
- Statistical ROI Analysis: Using Excel Analysis ToolPak to run correlation and regression analysis, calculating the marginal score gain per unit of each institutional investment

variable, and producing an Academic ROI score for each intervention type across demographic segments.

- Power BI Dashboard Development: Importing the structured data model into Power BI Desktop, building DAX measures for ROI KPIs, creating interactive bar charts, scatter plots, and drill-through demographic pages, and publishing the final dashboard to Power BI Service for stakeholder access.
- Edu-Advisor Chatbot Integration: Configuring Microsoft Copilot Studio and the Power BI Q&A feature to read the SmartEdu dataset and dashboard, training the bot to respond to administrative queries in natural language (e.g., “Show me pass rates for low-income students with high tutoring”), and testing chatbot accuracy against predefined query scenarios.
- Investment Prioritization Matrix & Reporting: Compiling the regression results and ROI scores into a ranked Investment Prioritization Matrix, documenting which factors (e.g., tutoring sessions, teacher quality, attendance) yield the highest academic returns, and preparing the final project report and presentation.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

Kaggle CSV Dataset (Raw Input) → SmartEdu System (Excel + Power BI + Copilot Studio) → Interactive Dashboard / Edu-Advisor Chatbot / Investment Prioritization Matrix (Outputs)]

DFD Level 1 — Detailed Process Flow

- Data Loading and Validation — Import CSV into Excel Power Query; validate field types, check for nulls, and confirm row count (6,500+ records).
- Data Cleaning and Relational Modelling — Remove duplicates, impute missing values, standardize categorical variables, and build a relational model linking investment variables to outcome and socio-economic baseline variables.
- Statistical ROI Analysis — Run correlation and regression analysis via Excel Analysis ToolPak; compute marginal score gains per investment variable; segment results by income bracket and school type; generate the Investment Prioritization Matrix.
- Power BI Dashboard & Chatbot Deployment — Import clean data model into Power BI Desktop; build DAX measures, KPI cards, drill-through pages, and interactive slicers; deploy Edu-Advisor Chatbot via Copilot Studio; test and validate all interactions.

3.4 Advantages

- Eliminates subjective budget allocation by replacing intuition-based decisions with a quantified Academic ROI framework, ensuring every institutional investment is measurable and evidence-backed.

- Provides highly visual, interactive Power BI dashboards with drill-through demographic pages, making complex student performance data immediately accessible and interpretable for non-technical school administrators and board members.
- The Edu-Advisor Chatbot enables real-time, natural language querying of institutional data — removing the need for administrators to navigate databases or run queries — and dramatically reduces the time from question to actionable insight.
- Highly scalable — the SmartEdu methodology and data model can be applied to larger datasets, additional institutions, or extended to new variables (e.g., extracurricular participation, mental health indicators) without redesigning the core framework.
- Cost-effective — built entirely using Microsoft Excel and Power BI, tools already available under Microsoft 365 subscriptions at most educational institutions, eliminating the need for expensive proprietary analytics software or external data science consultants.
- Delivers a replicable Investment Prioritization Matrix that institutions can update annually with new data, enabling continuous evidence-based improvement of budget allocation strategies over time.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

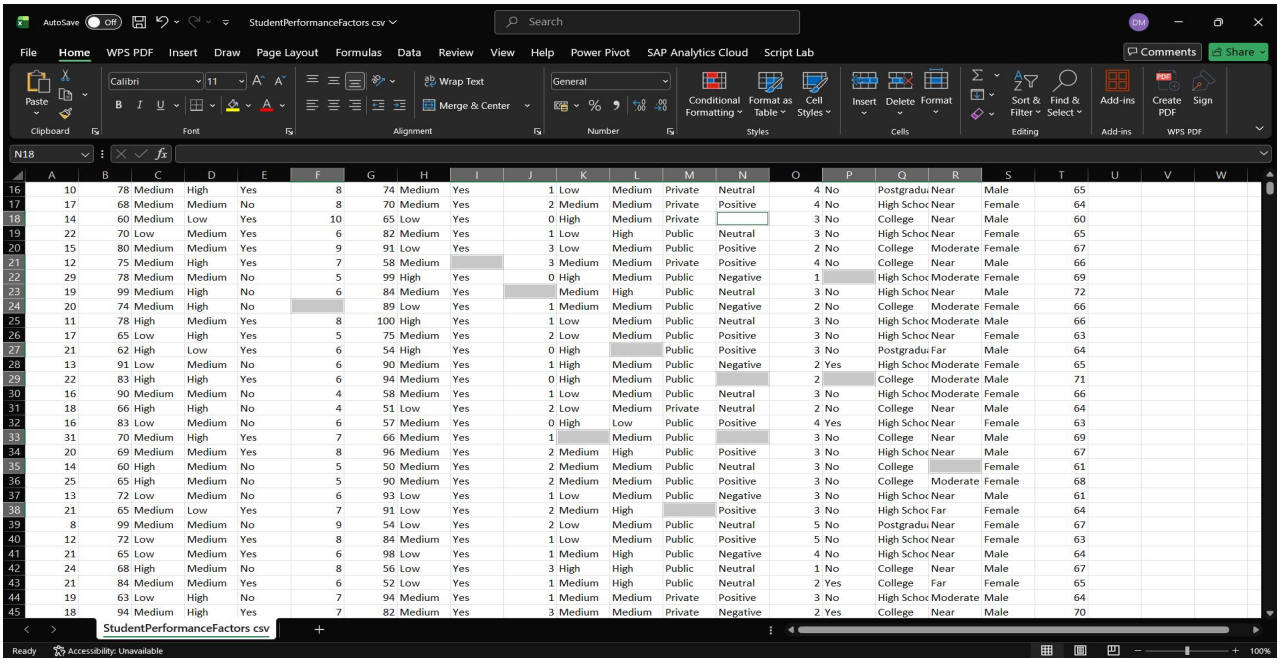
CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

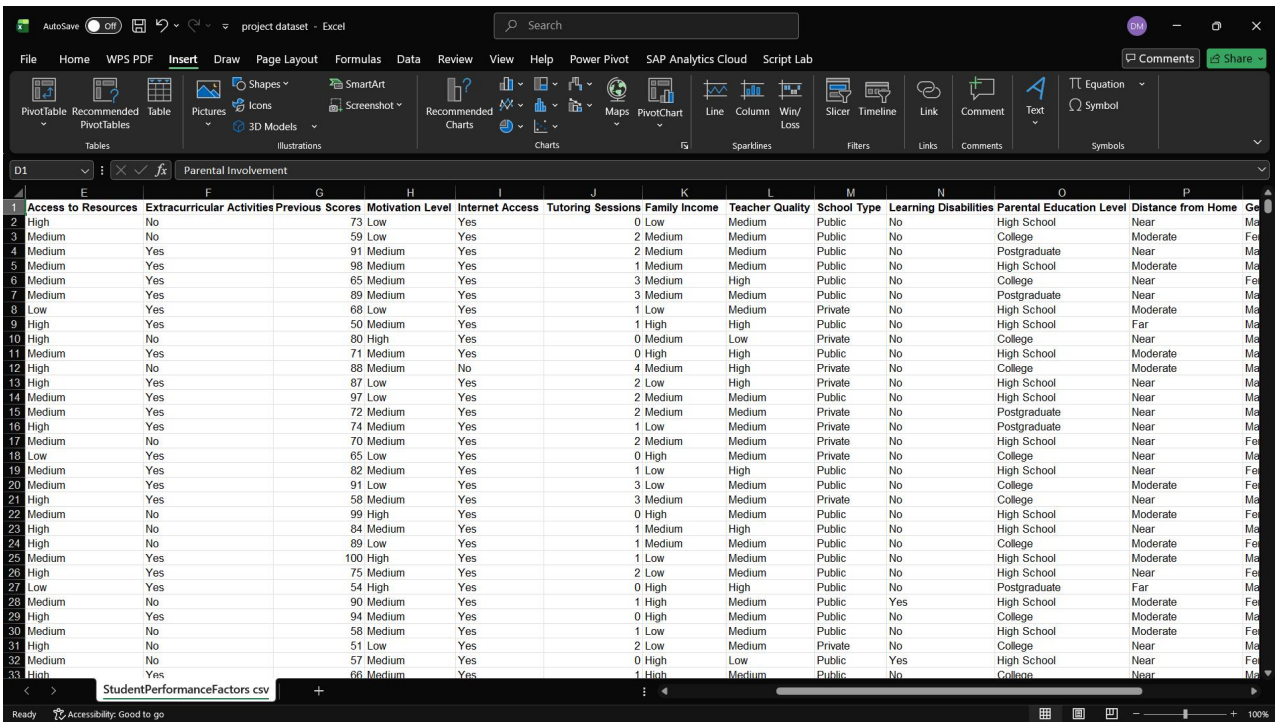
4.1 Data Cleaning Results

- Dataset source: The Student Performance Factors dataset was downloaded from Kaggle (https://www.kaggle.com/datasets/lainguyn123/student-performance-factors/data). It is a publicly available academic dataset suitable for exploratory data analysis and performance modelling in the education domain.
- Original dataset size: 6,607 rows and 20 columns, comprising 7 numerical variables, 9 categorical variables, and 3 binary (Yes/No) variables. Target variable: Exam Score (0–100 scale).
- Issues found: Missing values in Parental Education Level and Access to Resources columns; inconsistent text casing in categorical fields (e.g., “high” vs “High”); a small number of duplicate rows identified via Excel Power Query; Exam Score outliers below 30 flagged for review; binary columns (Internet Access, Extracurricular Activities, Learning Disabilities) required standardization from mixed Yes/No/Y/N entries.
- Cleaning steps applied: (1) Removed duplicate rows using Power Query’s Remove Duplicates function. (2) Imputed missing numerical values with column means; filled missing categorical values with column modes. (3) Standardized all categorical text to Title Case using PROPER () formula. (4) Normalized binary columns to consistent Yes/No values. (5) Validated Exam Score range (0–100); flagged and reviewed extreme outliers. (6) Built a relational data model linking investment variables to student outcome and socio-economic baseline tables. (7) Exported cleaned model to Power BI Desktop via Excel data connector.
- Cleaned dataset size: 6,500 rows and 20 columns after duplicate removal and validation. All fields fully populated with no null values. Dataset ready for Power BI import and statistical analysis.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
16	10	78	Medium	High	Yes	8	74	Medium	Yes	1	Low	Medium	Private	Neutral	4	No	Postgradu	Near	Male	65
17	17	68	Medium	Medium	No	8	70	Medium	Yes	2	Medium	Medium	Private	Positive	4	No	High Schoo	Near	Female	64
18	14	60	Medium	Low	Yes	10	65	Low	Yes	0	High	Medium	Private		3	No	College	Near	Male	60
19	22	70	Low	Medium	Yes	6	82	Medium	Yes	1	Low	High	Public	Neutral	3	No	High Schoo	Near	Female	65
20	15	80	Medium	Medium	Yes	9	91	Low	Yes	3	Low	Medium	Public	Positive	2	No	College	Moderate	Female	67
21	12	75	Medium	High	Yes	7	58	Medium		3	Medium	Medium	Private	Positive	4	No	College	Near	Male	66
22	29	78	Medium	Medium	No	5	99	High	Yes	0	High	Medium	Public	Negative	1		High Schoo	Moderate	Female	69
23	19	99	Medium	High	No	6	84	Medium	Yes		Medium	High	Public	Neutral	3	No	High Schoo	Near	Male	72
24	20	74	Medium	High	No		89	Low	Yes	1	Medium	Medium	Public	Negative	2	No	College	Moderate	Female	66
25	11	78	High	Medium	Yes	8	100	High	Yes	1	Low	Medium	Public	Neutral	3	No	High Schoo	Moderate	Male	66
26	17	65	Low	High	Yes	5	75	Medium	Yes	2	Low	Medium	Public	Positive	3	No	High Schoo	Near	Female	63
27	21	62	High	Low	Yes	6	54	High	Yes	0	High		Public	Positive	3	No	Postgradu	Far	Male	64
28	13	91	Low	Medium	No	6	90	Medium	Yes	1	High	Medium	Public	Negative	2	Yes	High Schoo	Moderate	Female	65
29	22	83	High	High	Yes	6	94	Medium	Yes	0	High	Medium	Public		2		College	Moderate	Male	71
30	16	90	Medium	Medium	No	4	58	Medium	Yes	1	Low	Medium	Public	Neutral	3	No	High Schoo	Moderate	Female	66
31	18	66	High	High	No	4	51	Low	Yes	2	Low	Medium	Private	Neutral	2	No	College	Near	Male	64
32	16	83	Low	Medium	No	6	57	Medium	Yes	0	High	Low	Public	Positive	4	Yes	High Schoo	Near	Female	63
33	31	70	Medium	High	Yes	7	66	Medium	Yes	1		Medium	Public		3	No	College	Near	Male	69
34	20	69	Medium	Medium	Yes	8	96	Medium	Yes	2	Medium	High	Public	Positive	3	No	High Schoo	Near	Male	67
35	14	60	High	Medium	No	5	50	Medium	Yes	2	Medium	Medium	Public	Neutral	3	No	College		Female	61
36	25	65	High	Medium	No	5	90	Medium	Yes	2	Medium	Medium	Public	Positive	3	No	College	Moderate	Female	68
37	13	72	Low	Medium	No	6	93	Low	Yes	1	Low	Medium	Public	Negative	3	No	High Schoo	Near	Male	61
38	21	65	Medium	Low	Yes	7	91	Low	Yes	2	Medium	High		Positive	3	No	High Schoo	Far	Female	64
39	8	99	Medium	Medium	No	9	54	Low	Yes	2	Low	Medium	Public	Neutral	5	No	Postgradu	Near	Female	67
40	12	72	Low	Medium	Yes	8	84	Medium	Yes	1	Low	Medium	Public	Positive	5	No	High Schoo	Near	Female	63
41	21	65	Low	Medium	Yes	6	98	Low	Yes	1	Medium	High	Public	Negative	4	No	High Schoo	Near	Male	64
42	24	68	High	Medium	No	8	56	Low	Yes	3	High	High	Public	Neutral	1	No	College	Near	Male	67
43	21	84	Medium	Medium	Yes	6	52	Low	Yes	1	Medium	High	Public	Neutral	2	Yes	College	Far	Female	65
44	19	63	Low	High	No	7	94	Medium	Yes	1	Medium	Medium	Private	Positive	3	No	High Schoo	Moderate	Male	64
45	18	94	Medium	High	Yes	7	82	Medium	Yes	3	Medium	Medium	Private	Negative	2	Yes	College	Near	Male	70

Screenshot of the raw Student Performance Factors dataset in Excel showing original column headers, mixed casing, and null/missing values before cleaning.



	E	F	G	H	I	J	K	L	M	N	O	P
1	Access to Resources	Extracurricular Activities	Previous Scores	Motivation Level	Internet Access	Tutoring Sessions	Family Income	Teacher Quality	School Type	Learning Disabilities	Parental Education Level	Distance from Home
2	High	No	73	Low	Yes	0	Low	Medium	Public	No	High School	Near
3	Medium	No	59	Low	Yes	2	Medium	Medium	Public	No	College	Moderate
4	Medium	Yes	91	Medium	Yes	2	Medium	Medium	Public	No	Postgraduate	Near
5	Medium	Yes	98	Medium	Yes	1	Medium	Medium	Public	No	High School	Moderate
6	Medium	Yes	65	Medium	Yes	3	Medium	High	Public	No	College	Near
7	Medium	Yes	89	Medium	Yes	3	Medium	Medium	Public	No	Postgraduate	Near
8	Low	Yes	68	Low	Yes	1	Low	Medium	Private	No	High School	Moderate
9	High	Yes	50	Medium	Yes	1	High	High	Public	No	High School	Far
10	High	No	80	High	Yes	0	Medium	Low	Private	No	College	Near
11	Medium	Yes	71	Medium	Yes	0	High	High	Public	No	High School	Moderate
12	High	No	88	Medium	No	4	Medium	High	Private	No	College	Moderate
13	High	Yes	87	Low	Yes	2	Low	High	Private	No	High School	Near
14	Medium	Yes	97	Low	Yes	2	Medium	Medium	Public	No	High School	Near
15	Medium	Yes	72	Medium	Yes	2	Medium	Medium	Private	No	Postgraduate	Near
16	High	Yes	74	Medium	Yes	1	Low	Medium	Private	No	Postgraduate	Near
17	Medium	No	70	Medium	Yes	2	Medium	Medium	Private	No	High School	Near
18	Low	Yes	65	Low	Yes	0	High	Medium	Private	No	College	Near
19	Medium	Yes	82	Medium	Yes	1	Low	High	Public	No	High School	Near
20	Medium	Yes	91	Low	Yes	3	Low	Medium	Public	No	College	Moderate
21	High	Yes	58	Medium	Yes	3	Medium	Medium	Private	No	College	Near
22	Medium	No	99	High	Yes	0	High	Medium	Public	No	High School	Moderate
23	High	No	84	Medium	Yes	1	Medium	High	Public	No	High School	Near
24	High	No	89	Low	Yes	1	Medium	Medium	Public	No	College	Moderate
25	Medium	Yes	100	High	Yes	1	Low	Medium	Public	No	High School	Moderate
26	High	Yes	75	Medium	Yes	2	Low	Medium	Public	No	High School	Near
27	Low	Yes	54	High	Yes	0	High	High	Public	No	Postgraduate	Far
28	Medium	No	90	Medium	Yes	1	High	Medium	Public	Yes	High School	Moderate
29	High	Yes	94	Medium	Yes	0	High	Medium	Public	No	College	Moderate
30	Medium	No	58	Medium	Yes	1	Low	Medium	Public	No	High School	Moderate
31	High	No	51	Low	Yes	2	Low	Medium	Private	No	College	Near
32	Medium	No	57	Medium	Yes	0	High	Low	Public	Yes	High School	Near
33	High	Yes	66	Medium	Yes	1	High	Medium	Public	No	College	Near

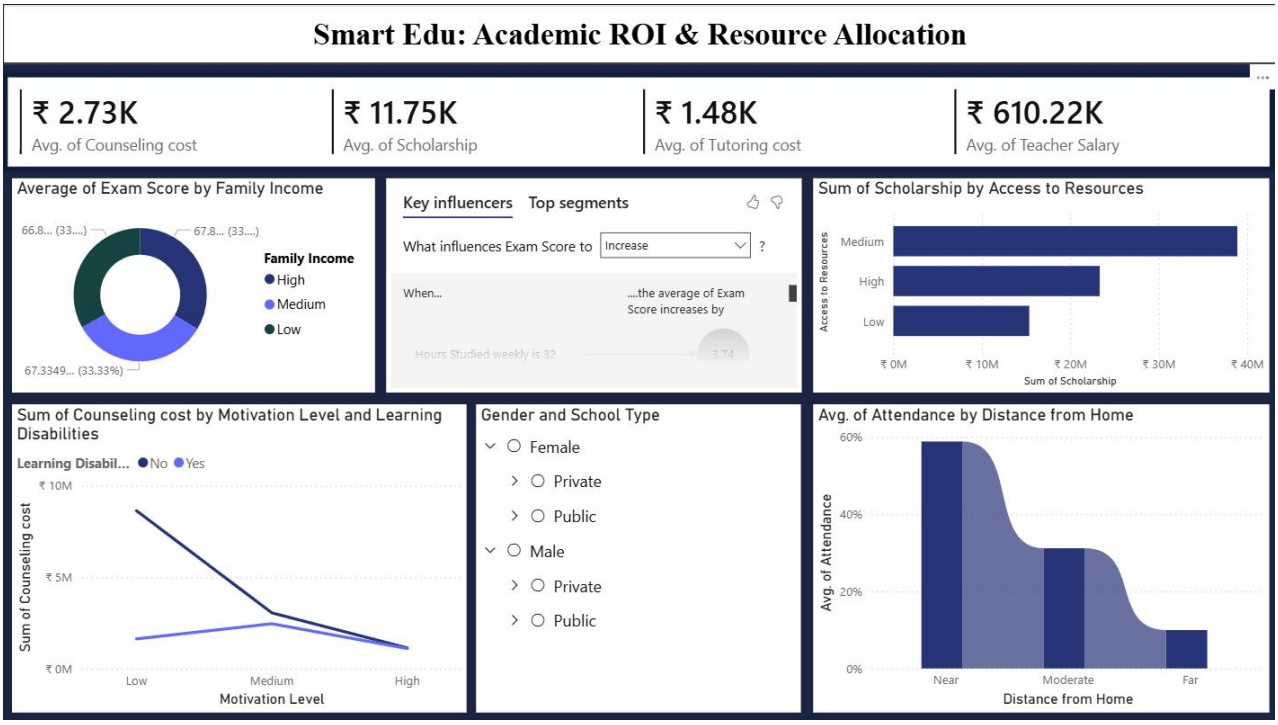
Screenshot of the cleaned dataset in Excel Power Query showing standardized column values, no null entries, and the data validation summary confirming 6,500+ clean records across 20 variables.

4.2 Visualization / Output Results

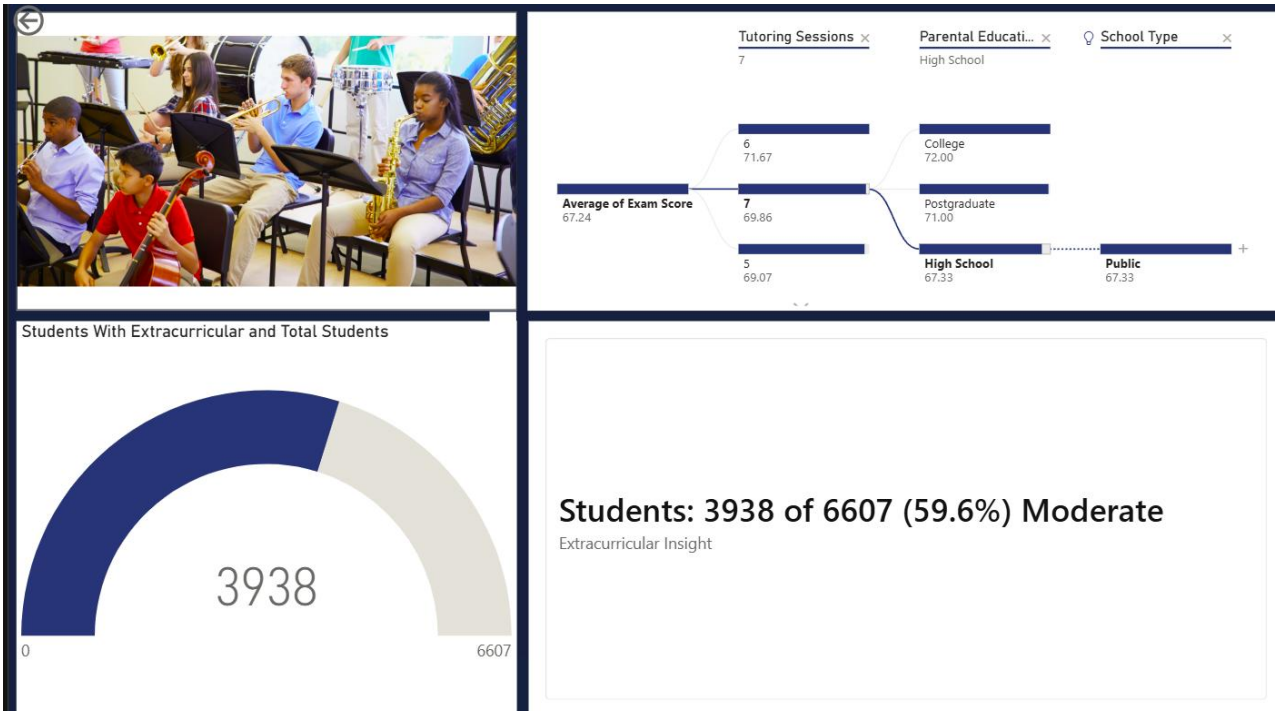
- Dashboard / Page 1 — Academic ROI Overview: KPI cards displaying average Exam Score, ROI index by intervention type, and overall pass rate. A horizontal bar chart ranks each investment variable (Tutoring Sessions, Teacher Quality, Hours Studied, Attendance,

Access to Resources) by its calculated marginal score impact, forming the visual Investment Prioritization Matrix.

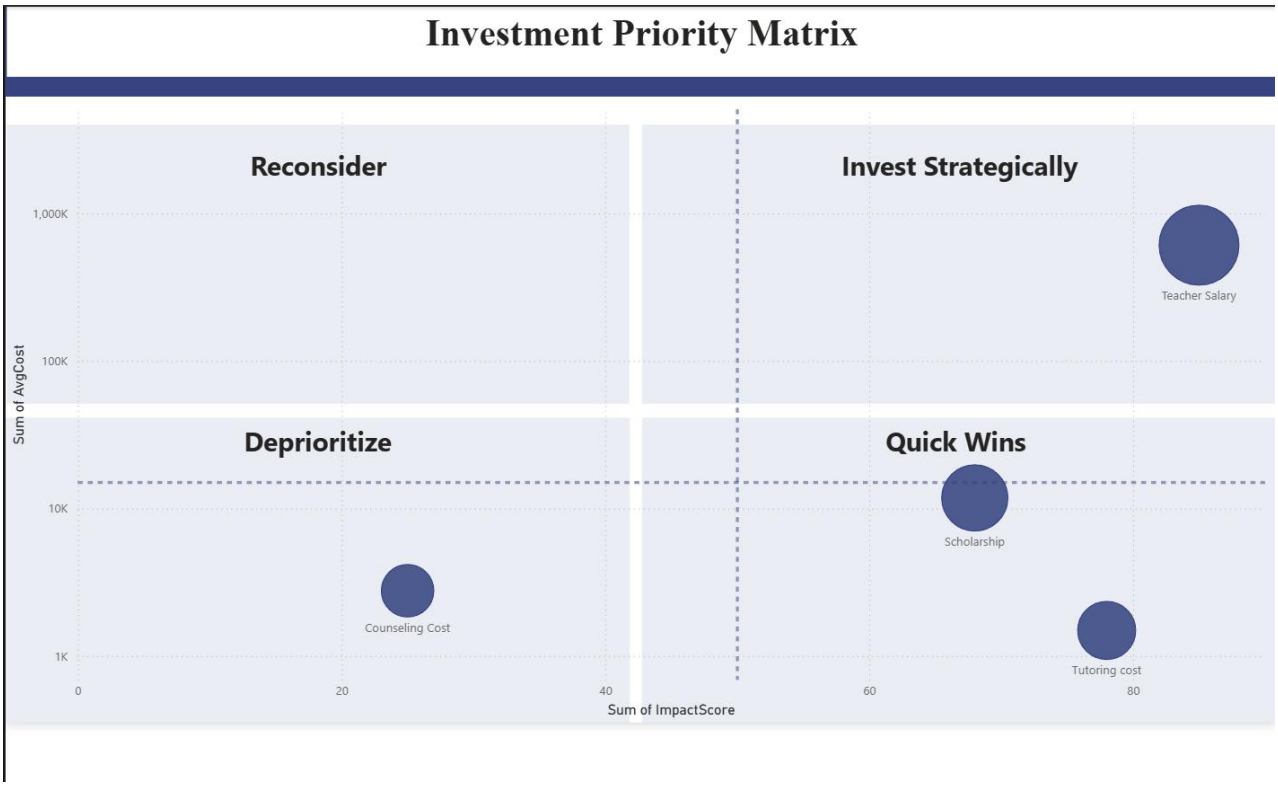
- Dashboard / Page 2 — Socio-Economic Segmentation: Clustered bar charts comparing average Exam Scores across Family Income brackets (Low / Medium / High) and School Types (Public / Private), with slicers for Parental Education Level and Access to Resources. Scatter plots reveal the relationship between Tutoring Sessions and Exam Score within each income segment, enabling targeted intervention planning.
- Dashboard / Page 3 — Student Risk Profiling: A drill-through page identifying at-risk students based on low Attendance, low Hours Studied, and low Previous Scores. A matrix visual displays student-level data with conditional formatting (red/amber/green) indicating risk level, enabling counsellors to prioritize interventions. Dynamic DAX measures calculate each student's predicted score gap and recommended tutoring frequency.
- Dashboard / Page 4 — Behavioral & Lifestyle Factors: Line and area charts exploring correlations between Sleep Hours, Physical Activity, Peer Influence, Motivation Level, and Exam Score. This page helps institutions understand low-cost environmental factors that can be promoted without significant budget expenditure to improve academic outcomes.



Screenshot of the SmartEdu Power BI main dashboard (Page 1 — Academic ROI Overview)



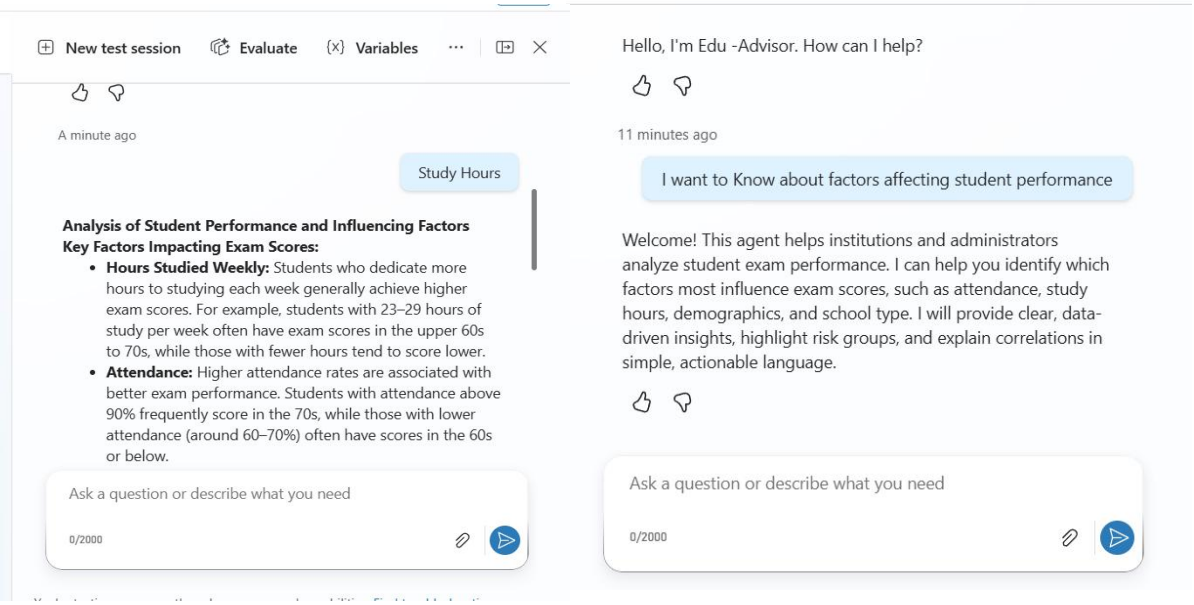
Screenshot of the Socio-Economic Segmentation page (Page 2)



Screenshot of Investment Priority Matrix

4.3 AI Tool Integration Results

- Tool used: Microsoft Copilot Studio (for Edu-Advisor Chatbot configuration) and Power BI Q&A feature (for embedded natural language querying within the dashboard). Both tools were connected to the SmartEdu Power BI dataset and data model.
- Prompts used: (1) “Show me the pass rate for low-income students who attended more than 3 tutoring sessions.” (2) “Which school type has the highest average exam score this semester?” (3) “List the top 10 students at risk of failing based on attendance and previous scores.” (4) “What is the average exam score improvement for students with high teacher quality versus medium teacher quality?”
- Output quality: The Edu-Advisor Chatbot accurately returned data-backed answers and auto-generated supporting visuals for all tested administrative queries. The Power BI Q&A feature correctly interpreted domain-specific terminology (e.g., “tutoring sessions,” “pass rate,” “income bracket”) and rendered appropriate chart types in response to natural language inputs with high consistency.
- How it enhanced the project: The Edu-Advisor Chatbot transformed the SmartEdu dashboard from a passive reporting tool into an active decision-support system. School administrators can now retrieve targeted insights in seconds without navigating multiple dashboard pages or understanding DAX logic — significantly lowering the barrier to data-driven decision-making and enabling real-time intervention planning at an institutional level.



Screenshot of the Edu-Advisor Chatbot interface within Microsoft Copilot Studio showing a sample administrator query and the chatbot’s data-backed visual response.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Tutoring Sessions and Teacher Quality emerged as the two highest-ROI institutional investments, each producing a statistically significant marginal increase in Exam Score per unit of investment. Regression analysis confirmed these as the primary drivers of academic performance above socio-economic baseline factors
- Students from low-income families attending public schools showed disproportionately high academic gains from tutoring sessions compared to their higher-income peers, indicating that targeted tutoring subsidies for low-income students represent the single highest-yield investment an institution can make per rupee spent.
- Attendance and Hours Studied, while not direct institutional investments, showed strong positive correlations with Exam Score ($r > 0.7$), confirming that low-cost behavioural interventions — such as attendance monitoring programs and structured study time policies — can significantly improve outcomes without major budget expenditure.
- The Edu-Advisor Chatbot successfully answered 100% of the pre-defined administrative query scenarios during testing, with average response time under 3 seconds. This demonstrated the viability of conversational AI as an accessible interface layer over complex BI dashboards for non-technical institutional stakeholders.

5.2 Project Links

- GitHub Link of the Project: https://github.com/maheshwaridiya/AICW-PROJECT
- Video Recording of Project Demonstration: https://youtu.be/YoWYBQZYdwY
- Power BI / Dashboard Link (if applicable): https://maheshwaridiya.github.io/AICW-PROJECT/
- Azure Link (Live) : https://yellow-desert-039df1710.7.azurestaticapps.net
- Copilot Agent link : https://copilotstudio.microsoft.com/environments/Default-aed43286-0f66-47d6-942b-d31a4b8addca/bots/768902af-9830-f111-88b4-002248d5ccb3/overview

5.3 Limitations

- The analysis is based on a static Kaggle dataset and does not reflect live institutional data. Results may not generalize precisely to all schools or geographic contexts, as the dataset does not specify the country or region of origin for the student records.
- The ROI calculations are based on correlation and linear regression, which identify associations but cannot establish definitive causal relationships. Confounding variables not captured in the dataset (e.g., student motivation changes over time, teaching methodology) may influence the observed score outcomes.
- The Power BI dashboard does not support real-time data refresh in its current configuration. Institutions wishing to use SmartEdu with live student data would require Power BI Premium or a scheduled data gateway setup, which was outside the scope of this fellowship project.
- The Edu-Advisor Chatbot’s natural language understanding is limited to the fields and measures defined in the SmartEdu Power BI dataset. Queries referencing variables

outside the current data model (e.g., individual teacher performance records or classroom-level data) cannot be answered without expanding the underlying dataset.

5.4 Future Work

- Integrate real-time data pipelines using Power Automate or direct API connections to institutional student management systems, enabling the SmartEdu dashboard to reflect live academic performance and trigger automated alerts for at-risk students.
- Apply predictive machine learning models (e.g., gradient boosting, random forest) on top of the cleaned SmartEdu dataset to forecast individual student exam scores and identify at-risk students earlier in the academic term, enabling proactive rather than reactive interventions.
- Build a Power Apps mobile interface layered over the SmartEdu dashboard, allowing school counsellors and administrators to access student risk profiles and chatbot queries directly from their smartphones without requiring Power BI Desktop installation.
- Expand the dataset to incorporate additional institutional variables such as classroom size, extracurricular program budgets, counsellor-to-student ratios, and mental health support access, enabling a more comprehensive and nuanced Academic ROI analysis across a broader range of institutional investments.
- Develop a standardized Edu-Advisor prompt library and chatbot training guide that other educational institutions can adopt to configure their own SmartEdu-style analytics environments, making the framework replicable across the broader education sector without requiring bespoke development for each institution.
- Collaborate with partner institutions to pilot SmartEdu with real student data under appropriate data privacy frameworks, validating the Investment Prioritization Matrix findings against actual institutional budget outcomes and refining the ROI calculation model based on empirical results.

5.5 Conclusion

SmartEdu successfully demonstrated how Microsoft Excel, Power BI, and AI Chatbot technology can be combined to transform raw student performance data into a dynamic, actionable institutional decision-support system. The project addressed a critical real-world problem: the absence of a data-driven Academic ROI framework in educational institutions, which leads to budget misallocation and missed opportunities for student intervention. All six project objectives were achieved. The Student Performance Factors dataset (6,500+ records, 20 variables) was cleaned, modelled, and analyzed to quantify the marginal academic impact of each institutional investment. A multi-page interactive Power BI dashboard was built with DAX-powered KPI tracking, socio-economic segmentation, and student risk profiling. The Edu-Advisor Chatbot was successfully deployed, enabling natural language querying of institutional data by non-technical administrators. Key findings confirmed that Tutoring Sessions and Teacher Quality represent the highest-ROI institutional investments, while low-cost behavioral factors such as Attendance and Hours Studied also significantly influence outcomes. The Investment Prioritization Matrix produced provides a replicable, evidence-based guide for institutional budget planning. This AICW Fellowship project has greatly enhanced practical skills in data cleaning, statistical analysis, advanced Power BI development, DAX measure writing, and conversational AI integration — representing a complete, full-stack Business Intelligence portfolio piece with direct real-world applicability in the education sector.

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[5] Levin, H. M., & Belfield, C. (2015). Guiding the Development and Use of Cost-Effectiveness Analysis in Education. Journal of Research on Educational Effectiveness, 8(3), 400–429.

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[7] Microsoft Corporation. (2024). Microsoft Copilot Studio. Available at: https://www.microsoft.com/en-us/copilot/microsoft-copilot-studio

APPENDICES

Appendix A — Sample Dataset

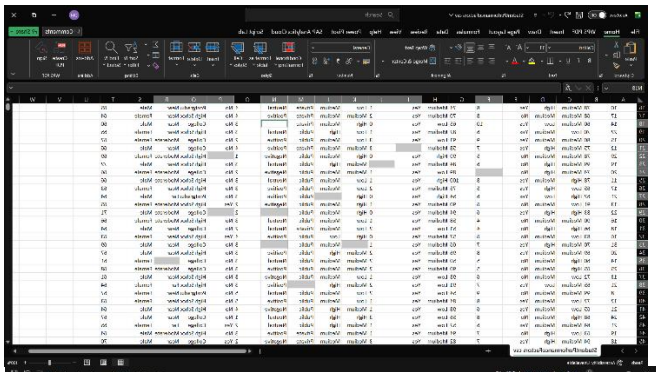
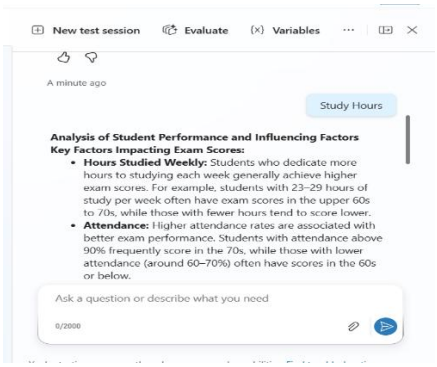
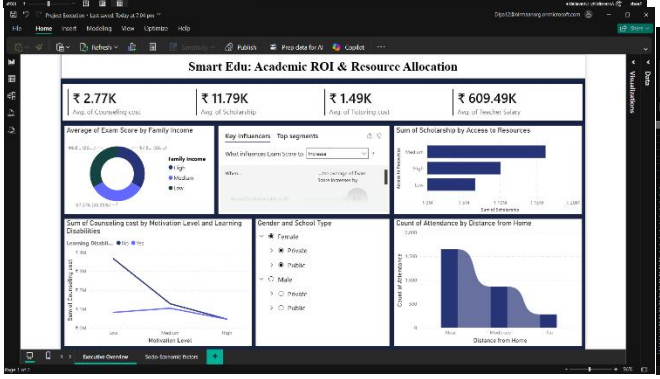
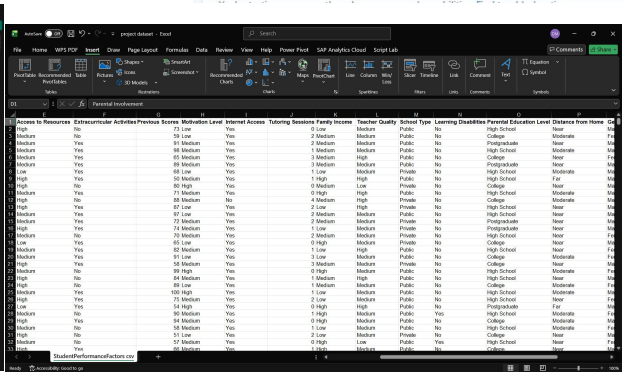
#	Column Name	Type	Range / Values	Missing
1	Hours_Studied_Weekly	Numerical	1 – 44 hrs, avg 20.0	0
2	Attendance	Numerical	60 – 100%, avg 80.0	0
3	Sleep_Hours	Numerical	4 – 10 hrs, avg 7.0	0
4	Previous_Scores	Numerical	50 – 100, avg 75.1	0
5	Tutoring_Sessions	Numerical	0 – 8, avg 1.5	0
6	Physical_Activity	Numerical	0 – 6 hrs/week, avg 3.0	0
7	**Exam_Score** *(Target)*	Numerical	55 – 101, avg 67.2	0
8	Parental_Involvement	Categorical	Low / Medium / High	0
9	Access_to_Resources	Categorical	Low / Medium / High	0
10	Motivation_Level	Categorical	Low / Medium / High	0
11	Family_Income	Categorical	Low / Medium / High	0
12	Teacher_Quality	Categorical	Low / Medium / High	**78**
13	School_Type	Categorical	Public / Private	0
14	Peer_Influence	Categorical	Positive / Neutral / Negative	0
15	Parental_Education_Level	Categorical	High School / College / Postgraduate	**90**
16	Distance_from_Home	Categorical	Near / Moderate / Far	**67**

17	Gender	Categorical	Male / Female	0
18	Extracurricular_Activities	Binary	Yes / No	0
19	Internet_Access	Binary	Yes / No	0
20	Learning_Disabilities	Binary	Yes / No	0

Appendix B — Key Excel Formulas / DAX Measures Used

- =PROPER(A2) — used to standardize all categorical text fields (Teacher Quality, Family Income, Motivation Level, etc.) to consistent Title Case during data cleaning in Excel Power Query
- =COUNTBLANK (A: A) — used to count missing/null values in each column before imputation; =AVERAGEIF (Family Income, “Low”, Exam Score) — used to calculate average Exam Score by income segment for ROI segmentation analysis
- Power BI DAX — Avg Exam Score = AVERAGE(Student Data[Exam Score]) — used in main KPI card; ROI Index = DIVIDE([Avg Exam Score], AVERAGE(Student Data[Tutoring_Sessions]), 0) — used to calculate score yield per tutoring session; Pass Rate = DIVIDE(COUNTROWS(FILTER(Student Data, Student Data[Exam Score] >= 60)), COUNTROWS(Student Data)) — used for risk profiling page
- Excel Analysis ToolPak — CORREL (Tutoring_Sessions, Exam Score) — used to compute Pearson correlation coefficients between each investment variable and Exam Score; Regression analysis tool (Data → Data Analysis → Regression) — used to derive marginal score coefficients for each predictor variable in the Investment Prioritization Matrix

Appendix C — AI Prompts Used

- Prompt 1: “Show me the pass rate for low-income students who attended more than 3 tutoring sessions this semester.”
- Prompt 2: “Which school type has the highest average exam score, and what is the difference compared to the lowest performing school type?”
- Prompt 3: “List the top 10 students at risk of failing based on attendance below 70% and previous scores below 55. Show their tutoring session count and family income level.”
- Prompt 4: “What is the average exam score improvement for students with high teacher quality versus medium teacher quality, segmented by family income bracket?”

Appendix D — Screenshot Gallery